

Attenuators

[Attenuators](#) are composed of multiple floating segments or one long flexible surface that is placed parallel to oncoming waves, therefore flexing with the rise and fall of the water and generating electricity through an internal turbine or generator.

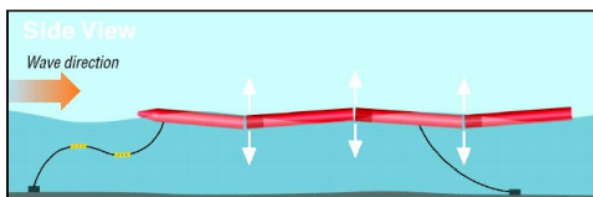


The Sea Wave Energy [Wave Line Magnet](#) attenuator

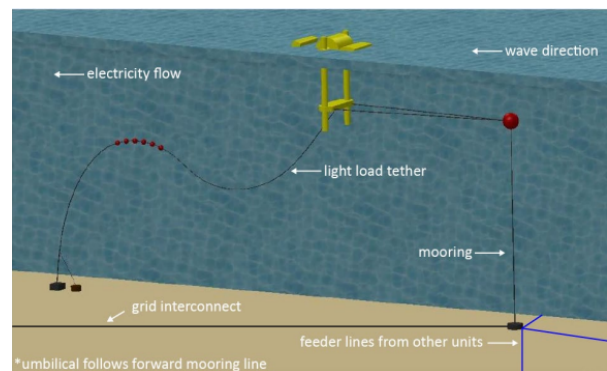


The [Waveberg](#) by Waveberg Development Ltd.

Mooring systems for these devices must meet the requirements of keeping the device in relative place while allowing flexible motion and maneuverability into the direction of oncoming waves. Therefore, they generally consist of at least one slack line. For example, the Pelamis P1 device included a slack line in the front allowing it to shift with wave direction. A similar configuration is used for the C-Power StingRAY, of which has a single-point mooring to an auxiliary buoy, allowing it to weathervane, with a slack cable limiting its range.

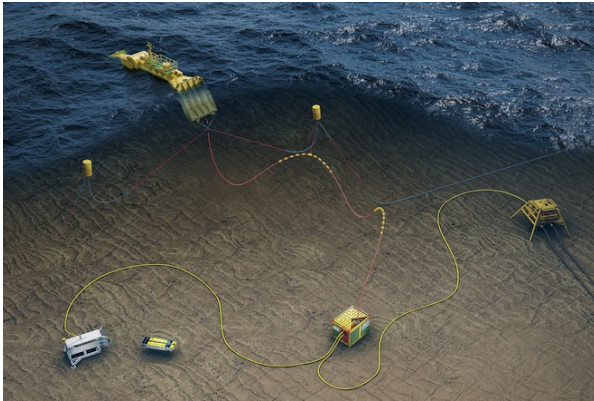


[Pelamis P1](#) and its mooring configuration.



[C-Power StingRAY](#) mooring configuration by Columbia Power Technologies Inc

These devices have ranged in size, with single hinged devices like the BlueStar 10 of which is 20m long, to the Pelamis P2 device of which was 180m long and 1,350 tons.



[Blue Star](#) by Mocean Energy



[Pelamis P2](#) by Pelamis Wave Power